

MATH 1203 BIT & CIS

DISCRETE MATHEMATICS

Lecture 4: RECURSION & RECURRENCES

Course Lecturer: M. KANYIKA

Applied Studies Department- MUST

9th March , 2026

Lesson Outcome

By the end of this lesson you should be able to:

- Define Recurrence relations.
- Solve problems involving recurrence relations
- Solve homogeneous linear recurrence relations problems .
- Solve problems involving second order homogeneous linear recurrence relations
- Solve General homogeneous linear recurrence relations problems .
- Solve non-homogeneous linear recurrence relations problems

INTRODUCTION

- I present to you third lecture of this course on the first topic, Recursion and Recurrence.
- In the previous lecture, you learnt about the Exclusion and inclusion principle as a counting technique.
- In this lecture, we will discuss about Recurrence relations.
- We will also discuss about homogeneous and non-homogeneous recurrence relation

Recursion & Recurrence Relations

Definition 1

A recurrence relation for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms $a_0, a_1, \dots, a_{n-1}$ for integers n with $n \geq n_0$ where n_0 is a nonnegative integer.

- A sequence is called a *solution* of recurrence relation if its terms satisfy the recurrence relation.
- The *Initial condition* is required to precede the first terms where relations take effect.

Recursion & Recurrence Relation CONT'D

Example 2

Let a_n be the sequence that satisfies the recurrence relation $a_n = a_{n-1} + 6$ for $n \geq 1$ given that $a_0 = 3$. Find $a_1, a_2, a_3 \dots a_{100}$

Recursion & Recurrence Relation CONT'D

solution

$$a_0 = 3$$

$$a_1 = a_0 + 6 = 3 + 6 = 9$$

$$a_2 = a_1 + 6 = 9 + 6 = 15$$

$$a_3 = a_2 + 6 = 15 + 6 = 21$$

$$a_4 = 3 + 4.6 = 27 \text{ (Using "a+nd".)}$$

$$\therefore a_{100} = a + n.d = 3 + (100 \times 6) = 603$$

Solving recurrence relation

- When solving recurrence relation we aim at finding a non-recursive formula to calculate a_n where the solution is called a closed formula.
- We can also use Iteration method involving substitution and back tracking methods to solve these recurrence relation.

Solving recurrence relation

Example 3

Let a_n be the sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n \geq 1$ with initial condition $a_0 = 2$. Use Iteration method involving substitution or back tracking method to solve for $a_1, a_2, a_3, a_4 \dots a_n$

Solution

- By Iteration method involving substitution we have

Solving recurrence relation

$$a_0 = 2$$

$$a_1 = 2 + 3$$

$$a_2 = a_1 + 3 = (2 + 3) + 3 = 2 + (3 + 3) = 2 + 2(3)$$

$$a_3 = a_2 + 3 = 2 + 2(3) + 3 = 2 + 3(3)$$

$$a_4 = 2 + 4(3)$$

$$\vdots$$

$$a_n = 2 + n(3)$$

$$a_n = 2 + 3n$$

Solving recurrence relation

- The use of back tracking is left for you as exercise.

Linear Recurrence Relations

- In linear recurrence , each term of a sequence is a Linear function of earlier terms in the sequence.

Definition 4

A linear homogeneous recurrence relation of degree K with constant coefficients is a recurrence relation of the form $a_n = C_1 a_{n-1} + C_2 a_{n-2} + \cdots + C_k a_{n-k}$ where $C_1, C_2, \cdots, C_k$ are real numbers and $C_k \neq 0$.

Linear Recurrence Relation CONT'D

- a_n is expressed in terms of the previous K terms of the sequence and therefore its degree is K .
- This recurrence include k initial conditions such that $a_0 = C_0$
- Consider a homogeneous second-order recurrence relation with constant coefficient which has the form $a_n = sa_{n-1} + ta_{n-2}$ or $a_n - sa_{n-1} - ta_{n-2} = 0$ where s and t are constants with $t \neq 0$.

Solving 2nd order homogeneous Linear Recurrence Relation CONT'D

- We associate the following quadratic polynomial with the above recurrence relation
$$\Delta(x) = x^2 - sx - t.$$
- This polynomial $\Delta(x)$ is called characteristics polynomial of the above recurrence relation and the roots are called *Characteristics roots*.

Solving 2nd order homogeneous Linear Recurrence Relation CONT'D

Theorem 5

Suppose the characteristic polynomial $\Delta(x) = x^2 - sx - t$ of the recurrence relation $a_n = sa_{n-1} + ta_{n-2}$ has distinct roots r_1 and r_2 . Then the general solution of the recurrence relation is denoted by $a_n = C_1 r_1^n + C_2 r_2^n$ where C_1 and C_2 are arbitrary constants.

Solving 2nd order homogeneous Linear Recurrence Relation CONT'D

Example 6

Consider the homogeneous recurrence relation $a_n = 2a_{n-1} + 3a_{n-2}$ with initial condition $a_0=1$, $a_1 = 2$.

- Find the general solution
- Find the unique solution with the given conditions.

Solving 2nd order homogeneous Linear Recurrence Relation CONT'D

Solution

a. Find the general solution

- First we find the characteristic polynomial $\Delta(x)$ and its roots r_1 and r_2 .

$$\begin{aligned}\Delta(x) &= x^2 - 2x - 3 = 0 \\ &= (x - 3)(x + 1) = 0\end{aligned}$$

- The roots $r_1=3$ and $r_2=-1$.
- Since the roots are distinct then by theorem 1 above the general solution is $a_n = C_1 3^n + C_2 (-1)^n$

Solving 2nd order homogeneous Linear Recurrence Relation CONT'D

Solution

b. Find the unique solution with the given conditions.

- Unique solution is obtained by finding C_1 and C_2 using initial conditions.
- For $n=0$ and $a_0 = 1$ we get $a_0 = C_1 3^0 + C_2 (-1)^0 = 1 \Rightarrow C_1 + C_2 = 1$
- For $n=1$ and $a_2 = 2$ we get $a_1 = C_1 3^1 + C_2 (-1)^1 = 2 \Rightarrow 3C_1 - C_2 = 2$
Solving the system of two equations in the unknown C_1 and C_2 yields $C_1 = \frac{3}{4}$

and $C_2 = \frac{1}{4}$. The unique solution of the given recurrence relation with the given initial conditions $a_0 = 1$ and $a_1 = 2$ is

$$a_n = \frac{3}{4} 3^n + \frac{1}{4} (-1)^n$$

Solving 2nd order homogeneous Linear Recurrence Relation CONT'D

Theorem 7

Suppose the characteristic polynomial $\Delta(x) = x^2 - sx - t$ of the recurrence relation $a_n = sa_{n-1} + ta_{n-2}$ has only one root r_0 (no distinct roots). Then the general solution of the recurrence relation where C_1 and C_2 are arbitrary constants is denoted by $a_n = C_1 r_0^n + C_2 n r_0^n$.

Solving 2nd order homogeneous Linear Recurrence Relation CONT'D

Example 8

Consider the following homogeneous recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ with initial condition $a_0=3$, $a_1 = 27$.

- a. Find the general solution
- b. Find the unique solution with the given conditions.

Solving 2nd order homogeneous Linear Recurrence Relation CONT'D

Solution

a. Find the general solution

- First we find the characteristic polynomial $\Delta(x)$ and its roots r_1 and r_2 .

$$\begin{aligned}\Delta(x) &= x^2 - 6x + 9 = 0 \\ &= (x - 3)(x - 3) = 0 \\ &= (x - 3)^2\end{aligned}$$

Thus $\Delta(x)$ has only one root $r_0=3$. Therefore by theorem 2 the general solution of the recurrence relation is $a_n = C_1 3^n + C_2 n 3^n$

Solving 2nd order homogeneous Linear Recurrence Relation CONT'D

Solution

b. Find the unique solution with the given conditions.

- The unique solution is obtained by finding C_1 and C_2 using the initial conditions.
- For $n = 1$ and $a_1 = 3$
- We get $C_1 3^1 + C_2(1)(3)^1 = 3$ or $3C_1 + 3C_2 = 3$.
- For $n = 2$ and $a_2 = 27$
- We get $C_1 3^2 + C_2(2)(3)^2 = 27$ or $9C_1 + 18C_2 = 27$.
- Solving the system of the two equations in the unknowns C_1 and C_2 yields:
 $C_1 = -1$ and $C_2 = 2$.
- The unique solution of the given recurrence relation with the given initial conditions

$$\begin{aligned}a_n &= -3^n + 2n3^n \\ &= 3^n(2n - 1)\end{aligned}$$

Non-homogeneous linear recurrence relations

- A linear non-homogeneous recurrence relation with constant coefficient is a recurrence relation of the form

$$a_n = C_1 a_{n-1} + C_2 a_{n-2} + C_3 a_{n-3} \cdots + C_k a_{n-k} + F(n)$$

where $C_1, C_2, \cdots C_k$ are real numbers and $F(n)$ is a function of n not identically zero.

- The recurrence relation $a_n = C_1 a_{n-1} + C_2 a_{n-2} + C_3 a_{n-3} \cdots + C_k a_{n-k}$ is called *associated homogeneous recurrence relation*.
- This recurrence includes k initial conditions.

$$a_0 = C_0 \quad a_1 = C_1 \cdots a_k = C_k$$

Non-homogeneous linear recurrence relations

- In non-homogeneous Linear recurrence relation with constant coefficient, every solution is the sum of the *particular solution* and the *associated linear homogeneous recurrence relations*
i.e. $a_n^{(h)} + a_n^{(p)}$

Theorem 9

If $\{a_n^{(p)}\}$ is a particular solution of non-homogeneous linear recurrence relation with constant coefficients $a_n = C_1 a_{n-1} + C_2 a_{n-2} + C_3 a_{n-3} \cdots + C_k a_{n-k} + F(n)$ then every solution is of the form $a_n^{(h)} + a_n^{(p)}$ where $a_n^{(h)}$ is a solution of the associated recurrence relation

Non-homogeneous linear recurrence relations CONT'D

Example 10

Find all solutions of the recurrence relation $y_n = 3y_{n-1} + 2n$ with initial condition $y_1 = 3$

Solution

We start by finding the associated homogeneous linear equation

$$\begin{aligned}y_n &= y_n^{(h)} + y_n^{(p)} \\y_n^{(h)} &= 3y_{n-1} \quad \text{since } r-3=0 \text{ then } r=3 \text{ (one root)} \\y_n^{(h)} &= C \cdot 3^n\end{aligned}$$

• The particular solution

Since $F(n) = 2n$ is a polynomial of degree n . A guess form (reasonable trial solution) is a linear function in n , say $p_n = Cn + d$, where c and d are constants.

- Then $y_n = 3y_{n-1} + 2n$ becomes

$$\begin{aligned} Cn + d &= 3[C(n-1) + d] + 2n \\ &= 3(Cn - c + d) + 2n \end{aligned}$$

$$Cn + d = 3Cn - 3c + 3d + 2n$$

$$0 = 3Cn - 3c + 3d + 2n - cn - d$$

$$0 = (2Cn + 2n)(-3c + 2d)$$

It follows that $Cn+d$ is a solution if and only if $2c+2=0$ or $2d-3c=0$. Hence $c=-1$ or $d=-\frac{3}{2}$. So the Particular solution is

$$P_n = Cn + d$$

$$y_n^{(P)} = -n - \frac{3}{2}$$

$$\begin{aligned} y_n &= y_n^{(h)} + y_n^{(p)} \\ &= C \cdot 3^n + \left(-n - \frac{3}{2}\right) \end{aligned}$$

We solve the constant term using the initial condition
 $y_1 = 3$ when $n=1$

$$Y_1 = C.3^1 + (-1 - \frac{3}{2}) = 3$$

$$3C = 3 + 1 + \frac{3}{2}$$

$$3C = \frac{11}{2}$$

$$C = \frac{11}{6}$$

Therefore the solution is $y_n = -n - \frac{3}{2} + (\frac{11}{6})3^n$

THANK YOU FOR BEING GOOD AUDIENCE!